SOFTWARE REQUIREMENT SPECIFICATION (SRS)

1. Introduction

The Human Resource Management System (HRMS) is an internal enterprise web-based application designed to automate and manage core HR operations such as travel management, expense reimbursement, employee engagement, games scheduling, organizational hierarchy visualization, and job referral workflows.

The system aims to reduce manual HR processes, improve transparency, and enhance employee engagement by providing a centralized digital platform accessible to Employees, Managers, and HR personnel.

I. User & Role Management Module

II. Travel & Expense Management Module

III. Social Achievements & Celebrations Module

IV. Games Scheduling Module

V. Organization Chart Module

VI. Job Listing, Sharing & Referral Module

2. Purpose

The primary purpose of the HRMS is to provide a secure, centralized, and efficient platform for managing internal HR operations.

The system aims to:

- Automate travel assignments and expense reimbursements
- Reduce manual document handling
- Improve employee engagement through social features
- Ensure fair allocation of shared organizational resources (games)
- Provide transparency in organizational hierarchy
- Simplify job sharing and referral processes

The system helps HR teams:

- Manage employee travel plans and documents
- Track and verify expenses
- Configure job postings and referral workflows
- View organization structure across departments

The system enables employees to:

- View assigned travels and submit expenses
- Upload and manage travel-related documents
- Participate in internal social activities
- Book game slots fairly
- Refer friends for job openings

3. Problem Statement

Organizations relying on manual or partially automated HR processes face challenges such as:

- Delays in travel communication and expense reimbursement
- Poor visibility of organizational hierarchy
- Unfair usage of shared recreational resources
- Inefficient job referral tracking
- Lack of employee engagement platforms

The HRMS solves these problems by digitizing and automating HR workflows with role-based access control and centralized data management.

4. Scope

The scope of the HRMS includes the development of a secure web-based system supporting the following roles:

- Employee
- Manager
- HR

Included Scope:

- Travel creation and assignment
- Expense submission and verification
- Social achievements feed with likes and comments
- Game slot scheduling with fairness rules
- Organization chart visualization
- Job listing, sharing, and referral management

Out of Scope (Future Enhancements):

- Payroll processing
- Performance appraisal
- Attendance management
- Automated payment disbursement

5. Audience and Users

I. **Employee:** Views travels, submits expenses, posts achievements, books game slots, refers jobs
II. **Manager:** View team member, travel documents, expense (read-only), organizational chart
III. **Hr:** Manager travels, expenses, documents, jobs, referrals, and organizational chart

6. Overview

I. Developer Requirement

Hardware Requirements	
Processor	Intel core i5 or higher
RAM	8 GB
Software Requirements	
Backend	ASP.NET Core
Database	SQL Server
Tools	Visual Studio, Visual Studio code, Postman
Browser	Latest versions of Chrome, Edge, Firefox

II. User Requirement

Hardware Requirements	
Processor	Intel core i3 or higher
RAM	4 GB
Software Requirements	
Operating System	Windows, Ios, Android
Browser	All Latest Browser

7. Acronyms and Abbreviations

- HRMS – Human Resource Management System
- API – Application Programming Interface
- DB – Database
- JWT – JSON Web Token
- RBAC – Role-Based Access Control

8. Functional Requirements

I. User & Role Management Module

a. Common Requirements:

- The system shall authenticate users using secure login credentials.
- The system shall enforce role-based access control.
- The system shall store encrypted passwords.

b. Employee:

- The employee shall be able to view and update profile information.
- The employee shall report to a manager and belong to a department.

c. Manager:

- The manager shall be able to view direct reports.
- The manager shall appear correctly in the organization chart.

d. Hr:

- HR shall be able to view all employees and departments.
- HR shall manage travel plans, expenses, jobs, and referrals.

II. Travel & Expense Management Module

a. Travel Assignment

- HR shall be able to create travel plans for one or more employees.

- The system shall notify employees via email and in-app notification.
- Each travel shall include dates, location, and expense limit.

b. Travel Documents

- The system shall allow HR and employees to upload documents.
- Each document shall store:
  - Travel ID
  - Employee ID
  - Uploaded by
  - Document type
  - File URL
  - Uploaded date
- Permissions:
  - HR: View/upload all documents
  - Employee: View/upload own documents
  - Manager: View team documents (read-only)

c. Expense Submission

- Expenses shall always be linked to a travel.
- Employees may submit expenses only:
  - After travel start date
  - Within 10 days of travel end date
- The system shall block late submissions.
- Multiple expenses per travel shall be supported.

d. Proof Upload & Verification

- Each expense must include at least one proof document.
- HR shall be able to approve or reject expenses.
- Rejection requires mandatory remarks.
- Status values: Draft, Submitted, Approved, Rejected

III. Social Achievements & Celebrations Module

a. Achievement Posts

- Employee Can Create Achievement posts with:
  - Title
  - Description
  - Tags
  - Visibility
- All User can view posts in reverse chronological order
- Users can like and comment on posts.

b. Moderation

- Employee can edit/delete their posts
- HR can delete inappropriate posts.
- System shall send a warning email on HR moderation

c. Birthday & Anniversaries

- The system shall auto-generate system posts for:
  - Employee Birthday
  - Work Anniversaries
- Auto-generate system posts shall be visible to all users

IV. Game Scheduling Module

a. Configuration

- HR/Admin can configure:
  - Operating Hours
  - Slot Durations
  - Max and Min Players per slot
- System shall generate daily slot automatically

b. Booking rules

- Employee can request slots for game
- One active booking per employee per day
- Fair scheduling shall be enforced.
- Cancellation shall assign next eligible player.

V. Organization Chart Module

- The system shall display the managerial chain of an employee
- The system shall show one level of direct reports
- Each node shall display:
  - Name
  - Designation
  - Profile image
- Clicking a node reloads the organization chart for that employee

VI. Job Listing, Sharing & Referral Module

a. Job listing

- The system shall list active job openings
- Employee can browse job details

b. Job Sharing:

- Employee can share job via email
- The system shall attach the job description document
- Sharing activity shall be logged

c. Job sharing

- Employee can refer friends with:
  - Name
  - Email
  - CV
  - Note
- The system shall email:
  - HR owner
  - Configured reviewers
- Referral status shall be tracked

9. Non-Functional Requirements:

I. Accessibility

- The system shall be accessible via modern web browsers.
- UI shall be responsive across devices.

II. Availability

- The system shall be available 24×7 except maintenance.
- Concurrent users shall be supported.

III. Performance

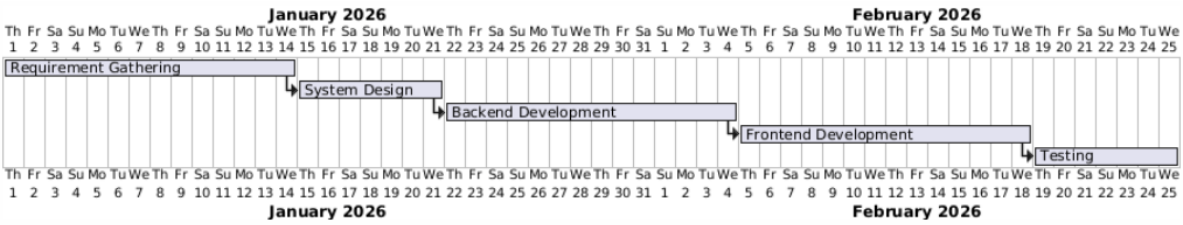
- Frequently accessed data shall be optimized.
- API responses shall be efficient.

IV. Security

- JWT-based authentication shall be enforced.
- Role-based authorization shall be applied.
- File uploads shall be validated.

V. Maintainability

- Modular architecture shall be followed.
- Database design shall support future expansion.

10. Other Information

I. Gantt Chart: